

DSC 257R - UNSUPERVISED LEARNING

CONDITIONAL INDEPENDENCE PROPERTIES OF MRFS

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Conditional Independence

Scenario: Alice and Bob don't know each other but drive home from work along the same road. Alice is late for dinner if she leaves work late, or if there is a lot of traffic that day; likewise with Bob.

Define three Boolean random variables:

- A : Is Alice late for dinner?
- B : Is Bob late for dinner?
- T : Is there a lot of traffic that day?

- 1 Is A independent of B ?
- 2 Is A independent of B given T ?

Conditional Independence: Definition

Let X, Y, Z be random variables defined on the same space.

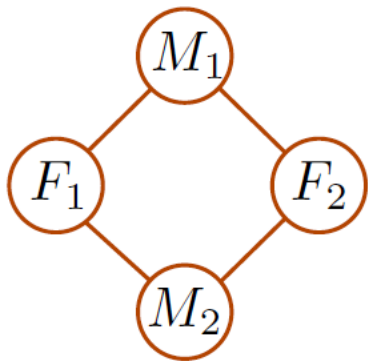
We say X is **conditionally independent of Y given Z** , denoted $X \perp\!\!\!\perp Y|Z$, if for all x, y, z ,

$$\Pr(X = x, Y = y|Z = z) = \Pr(X = x|Z = z) \Pr(Y = y|Z = z).$$

Example: Let X and Y be independent coin flips taking values $\{0,1\}$, and let $Z = X \wedge Y$. Is $X \perp\!\!\!\perp Y|Z$?

Back to Pearl's Disease Example

Boolean variables
(0/1): have disease?



F_1	M_1	$\Psi_{11}(F_1, M_1)$
0	0	100
0	1	20
1	0	20
1	1	50

F_2	M_1	$\Psi_{21}(F_2, M_1)$
0	0	200
0	1	10
1	0	100
1	1	50

F_1	M_2	$\Psi_{12}(F_1, M_2)$
0	0	100
0	1	20
1	0	20
1	1	50

F_2	M_2	$\Psi_{22}(F_2, M_2)$
0	0	100
0	1	20
1	0	20
1	1	1

$$P(F_1, F_2, M_1, M_2) = \frac{1}{Z} \Psi_{11}(F_1, M_1) \Psi_{12}(F_1, M_2) \Psi_{21}(F_2, M_1) \Psi_{22}(F_2, M_2)$$

What conditional independences are present in this distribution?

$$P(F_1, F_2, M_1, M_2) = \frac{1}{Z} \psi_{11}(F_1, M_1) \psi_{12}(F_1, M_2) \psi_{21}(F_2, M_1) \psi_{22}(F_2, M_2)$$

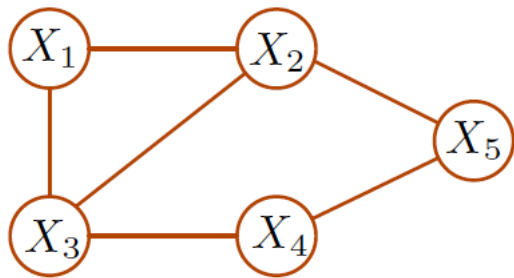
Recall: Markov Random Fields

Joint distribution over random variables $X_1, \dots, X_n$ given by:

- 1 An undirected graph with nodes $X_1, \dots, X_n$ and edges representing dependencies.
- 2 A distribution that factors over this graph:

$$P(X_1, \dots, X_n) = \frac{1}{Z} \prod_C \psi_C(\{X_i: i \in C\})$$

where the product is over maximal cliques in the graph, and the **clique potentials** ψ_C are positive-valued functions.



Functional form of $P(X_1, X_2, X_3, X_4, X_5)$

$$\frac{1}{Z} \psi_{123}(X_1, X_2, X_3) \psi_{25}(X_2, X_5) \psi_{34}(X_3, X_4) \psi_{45}(X_4, X_5)$$

Conditional Independence in MRFs

Let G be an undirected graph with nodes $X_1, \dots, X_n$.

Let $N_G(X_i)$ denote the neighbors of X_i in G .

Suppose the distribution $P(X_1, \dots, X_n)$ factors over this graph. Then it satisfies the following two properties:

① **Local independence property:** for any i ,

$$X_i \perp\!\!\!\perp \{X_j : j \neq i\} \mid N_G(X_i).$$

Easy proof: Algebraic manipulation of functional form of MRF.

② **Global independence property:** for any subsets of nodes S, T, U such that removing U separates S from T ,

$$X_S \perp\!\!\!\perp X_T \mid X_U.$$

The Hammersley-Clifford Theorem

Hammersley-Clifford Thm. Let P be a distribution on $(X_1, \dots, X_n)$ such that

- $P(x) > 0$ for all x , and
- P satisfies the local independence properties.

Then P is a Markov random field that factors over G .

Take-home message: From the (qualitative) conditional independence properties of a distribution, we can infer its precise functional form!